

The Average Pattern of Auroral Particle Precipitation, Its Associated Conductivity and Field Aligned Currents

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A series of statistical studies has been completed to determine the global pattern of auroral electron and ion precipitation and their resultant Hall and Pedersen conductivities as a function of geomagnetic activity, solar wind velocity, the orientation of the Interplanetary Magnetic Field (IMF) and season. In addition, studies were performed relating these global patterns in particle precipitation to similar global determinations of the average delta B vector produced by the auroral field aligned currents. The data for these studies were from instruments flown on the satellites of the Defense Meteorological Satellite Program (DMSP). In all cases the studies were performed by dividing the high latitude region into a series of spatial bins in Magnetic Local Time (MLT) and corrected geomagnetic latitude (CGL). One such matrix of spatial bins was created for each different value or range of values of chosen sort parameter. For geomagnetic activity the sort parameter was one of seven levels of Kp . For the IMF and solar wind velocity the sort parameters consisted of 30 paired ranges of the solar wind velocity and the B_z component of the IMF. A rough separation by the IMF B_y was performed by using the IMF sector structure and K_p together as the sort parameter. Seasonal separations were made with Kp and for time periods centered on the summer and winter solstices and the equinoxes. In all cases the large DMSP data sets were used to determine the average spectrum of precipitating electrons and ions and the average delta B vector for each spatial bin and for each sort parameter used. In this paper we review the results of these studies.

1. Introduction

Classically a system which exhibits high spatial and temporal variability is studied either by determining its behavior as a function of a single variable while all other variables are held constant or by determining its behavior while simultaneously measuring on the required spatial and temporal scales all the critical variables. Such approaches are in general not possible in geophysical systems where the systems can not be controlled and where the systems are so large that even with well instrumented satellites full measurements can not be made on temporal and spatial time scales appropriate to determining the details of the system. This is particularly true for determination of the global auroral zone processes where spatial scales for many processes are many thousands of kilometers in MLT and CGL and time scales vary from seconds to hours. In such a case one approach to studying the system is to take a large sample of one measurable available over the full spatial extent of the system and determine its average behavior as a function of different sort parameters that one might expect

to control the system in part. For appropriate sort parameters this can provide information on the underlying average configuration of the system.

A series of studies of this type have been completed using the data from instruments flown on the satellites of the Defense Meteorological Satellite Program (DMSP) (HARDY *et al.*, 1985; HARDY *et al.*, 1987; BRAUTIGAM *et al.*, 1988; HARDY *et al.*, 1989; RICH *et al.*, 1990; BRAUTIGAM *et al.*, 1990; HARDY *et al.*, 1990). In these studies the global pattern of electron and ion precipitation, the conductivities they produce, and their associated field aligned currents were determined as functions of geomagnetic activity, conditions in the interplanetary medium, and season. These studies provide important insights into the global processes producing the patterns, the regions in the magnetosphere to which the various parts of the oval connect, and the character of the interaction between the interplanetary medium and the magnetosphere. This work is an elaboration and extension of earlier work using other spacecraft data (REIDLER, 1972; MCDIARMID *et al.*, 1975; SPIRO *et al.*, 1982).

In this paper we review selected results from these studies. The paper is divided into four sections. Following the Introduction in Section 2 we describe the instrumentation providing the data and the methodology used to sort the data. In Section 3 the results of the studies are reviewed and Section 4 contains the summary and conclusions.

2. Instrumentation and Methodology

The data for these studies were provided by the SSJ/3 and SSJ/4 particle spectrometers and the SSM magnetometer all flown on the satellites of the Defense Meteorological Satellite Program. Both the SSJ/3 and SSJ/4 detectors consist of arrays of cylindrical electrostatic analyzers oriented on the spacecraft to measure continuously the flux of particles precipitating from the local zenith direction. The older SSJ/3 detectors measured electrons in 16 channels over the energy range from 50 eV to 20 keV. The newer SSJ/4 detector measures both electrons and ions in 20 channels over the extended range from 30 eV to 30 keV. Both detectors returned a complete energy spectrum once per second. (For more details see HARDY *et al.*, 1979; HARDY *et al.*, 1984). The SSM instrument is a triaxial fluxgate magnetometer, body mounted on the DMSP F7 satellite, that returns twenty vector measurements per second (For more detail see RICH *et al.*, 1984).

In general, at least two DMSP satellites are operated in orbit at all times. Both satellites are maintained in circular, polar, sun-synchronous orbits at an altitude of approximately 840 kilometers; one with its orbit plane in the dawn dusk meridian and the other in the 1030–2230 meridian. Because of the offset between the geographic and geomagnetic poles the satellites sweep over a wide range in MLT and CGL during the day covering the majority of the auroral region. The magnetometer and particle spectrometers are flown on the satellites to provide near real-time specification of the level of auroral activity and as such they are operated continuously with approximately 85% of the data recorded and returned to the ground. As such the instruments provide an extremely large data set (many tens of millions of measurements) that is particularly well suited for large statistical studies.

In all of the studies conducted, the high latitude region was divided into a series of spatial bins in MLT and CGL. The bins were defined by making 48, one-half hour divisions in MLT and 30 divisions in CGL. The CGL divisions were at 2 degree increments between 50 and 60 and 80 and 90 degrees and at one degree increments from 60 to 80 degrees. One such matrix of spatial bins was created for each different value or set of values of the sort parameters. For our earliest studies the sort parameter was Kp . For these studies one spatial

matrix was defined for each whole value of K_p from 0 to 5, i.e., one matrix for $K_p = 0$ and $0+$, one matrix for $K_p = 1-$, 1 , $1+$ etc. One final matrix was defined for all $K_p \geq 6-$ giving a total of 7 divisions. To examine seasonal variations we separated the data not only by K_p as above but also by three 90-day blocks in time centered on the winter and summer solstices and the equinoxes. To study the effect of the interplanetary medium we used as sort parameters paired values of the IMF B_z component (in geocentric solar magnetospheric coordinates) and V_{sw} . The matrix of sorting parameters had six divisions in B_z and five in V_{sw} . The divisions for B_z were -10 to -3 nT, -3 to -1.5 nT, -1.5 to 0 nT, 0 to $+1.5$ nT, $+1.5$ to $+3$ nT, and $+3$ to $+10$ nT. The divisions for solar wind velocity (V_{sw}) were 200 to 375 km/s, 375 to 450 km/s, 450 to 525 km/s, 525 to 625 km/s, and 625 to 800 km/s. This gave a matrix of thirty paired ranges of interplanetary parameters. Hourly averages of the IMF and V_{sw} provided by the National Space Science Data Center (COUZENS and KING, 1986, 1989) were used in the sort.

In all such studies the average electron and ion flux was determined in each of the energy channels of the SSJ/3 and SSJ/4 detectors in each spatial bin for each single or pair of sort parameters. The average spectra defined in this way were then integrated to determine the average integral energy flux (in units of keV/cm² sec ster), the average integral number flux (in units of particles/cm² sec ster) and average energy (in units of keV). The ion and electron spectra were extrapolated to higher energy for those latitudes and sort parameters where the average spectra indicated that a significant portion of the precipitating flux lay beyond the upper energy limit of the detector. The Hall and Pedersen conductivities produced by the precipitating electrons were calculated using the relationship between the integral energy flux and average energy and these conductivities developed by SPIRO *et al.* (1982) and ROBINSON *et al.* (1987). For more details on the binning techniques see HARDY *et al.* (1985), HARDY *et al.* (1989), and BRAUTIGAM *et al.* (1990).

For the determination of the average delta B vectors and their relationship to the precipitation pattern the same spatial binning was used. In this case the data were sorted by K_p , by season for time periods centered on the solstices and equinoxes, and by the IMF towards and away sector structure. The components of the magnetic field as measured by the SSM were fit to a baseline and the delta B deviations from that baseline determined. These were then used to determine the average delta B vector in corrected geomagnetic coordinates in the plane orthogonal to the magnetic field direction for each spatial bin and for each set of sort parameters. Similarly the average electron and ion spectra were determined in each spatial bin and for the same sets of sort parameters. For more details on the techniques see RICH *et al.* (1990).

3. Results

In this section we illustrate the results obtained from the series of statistical studies conducted with the DMSP data. Specifically, we will concentrate on three aspects of these works: 1) The general global pattern of auroral electron and ion precipitation and how they compare. 2) The variation in the global energy and particle input from auroral particles as a function of conditions in the interplanetary medium. 3) The global pattern of delta B 's and their relationship to the pattern of auroral particle precipitation.

3.1 Auroral electron and ion precipitation patterns and their resulting conductivities

In Figs. 1 and 2 we illustrate the results from the sorting of the particle data by Kp . In the figures we have plotted in a polar, color-spectrogram format, the global pattern of integral number flux, integral energy flux and average energy for both electrons and ions for $Kp = 3$ and the Hall and Pedersen conductivities for the same Kp produced by the precipitating electrons. Only the $Kp = 3$ case is presented since it is broadly representative of the general pattern of precipitation and conductivities seen for both higher and lower levels of geomagnetic activity. The principal change observed with activity is an increase in the general level of precipitation and in the size of the oval while the overall pattern within the oval is approximately maintained.

The global patterns for the integral energy flux for electrons and ions are similar. For both species the precipitation occurs within an approximately annular region with a circular equatorward boundary. For both species the highest levels of precipitation occur in c-shaped or crescent regions. The regions of higher flux are located towards the nightside with a uniform decrease towards noon. In both cases these regions of higher flux are symmetric about a meridian offset from midnight. For electrons the axis of symmetry of the pattern is offset towards a meridian post-midnight and post-noon while for ions the axis is offset towards a meridian pre-midnight and pre-noon. Indeed, the general pattern for electrons and ions appear to be approximate mirror images reflected about the noon-midnight meridian. This behavior is generally consistent with the motion of particles in the inner magnetosphere where the hot electrons should be transported towards the morningside flank of the magnetosphere and the hot ions towards the dusk flank. For both the electrons and ions there is a very large decrease in the average level of precipitation observed at the nightside maximum and at noon reflecting the decrease of the particle flux on a field line as it is transported and/or a change in the strength of the diffusion process with MLT that scatters particles into the atmospheric loss cone.

The largest difference in the integral energy flux patterns for electrons and ions is seen in the overall level of flux. In general, the electron integral energy flux exceeds that for the ions by approximately a factor of ten. The exception to this is found on the evening side of the oval. This is shown in Fig. 3 where we have plotted integral energy flux for electrons and ions along the dawn-dusk meridian. We note that on the evening side of the oval towards the equatorward edge of the precipitation region the ion integral energy flux equals or exceeds that for electrons while on the morningside the ion level reaches values equal to no more than a few percent of the electron level. Integrating over latitude on the evening side the contribution from ions can be as much as 50% that of the electrons. This behavior is found at all levels of activity. We have also found that in the region of the evening side oval where the ion integral energy flux is high, the ion average energy is as much as ten times the average energy of the electron average energy. BASU *et al.* (1987) have shown that such hot ion spectra will produce a peak in ionization at approximately the same altitude as the cooler electrons. This means that on the evening side of the oval the ions make a significant contribution to the ionospheric conductivity and their effect must be taken into account in determining the overall dynamics of this region.

For the integral number flux the global patterns for ions and electrons differ more substantially; particularly on the dayside. For both species the maximum value of the integral number flux is found on the dayside with a weaker secondary maximum on the nightside. The intensity of the number flux on the dayside remains approximately constant with increasing Kp while the flux on the nightside increases with increasing Kp exceeding

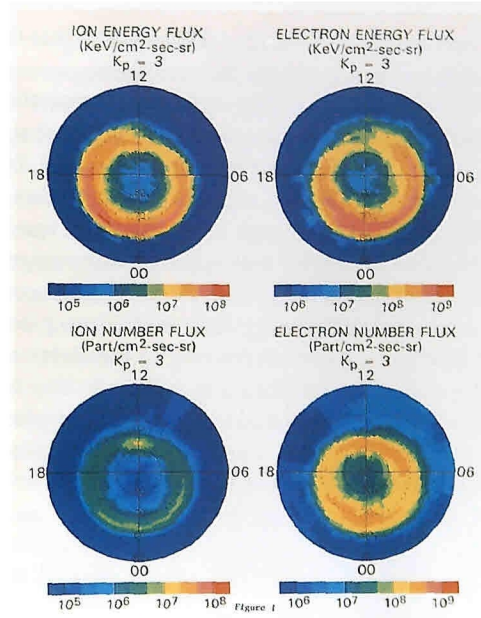


Fig. 1. Polar color spectrograms of the global pattern of precipitating electrons and ions for $K_p = 3$. The top two plot show the pattern of integral energy flux ($\text{keV}/\text{cm}^2 \text{ sec ster}$) and the bottom two the pattern for the integral number flux ($\text{particles}/\text{cm}^2 \text{ sec ster}$) for ion (left) and electrons (right). The data are plotted in a Magnetic Local Time/Corrected Geomagnetic Latitude coordinate system.

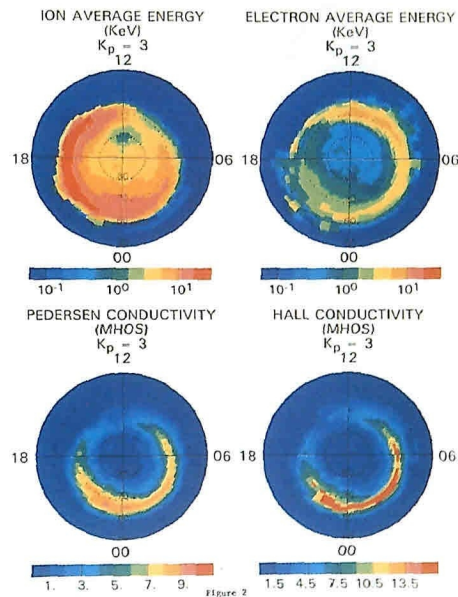


Fig. 2. Polar color spectrograms in the same format as Fig. 1. The top two plots show the electron and ion patterns of the average energy. The bottom two plots show the pattern of the height integrated Hall and Pedersen conductivities produced by the precipitating electrons.

the maximum dayside level for higher Kp levels. This reflects that the dayside precipitation associated with the cusp/cleft and connecting to the various magnetospheric boundary regions remains constant with increasing Kp while the nightside precipitation originating from the plasma sheet increases. The two patterns differ most markedly in that for electrons the region of high flux occurs over a wide cleft-shaped region limited in CGL but extending in MLT to the dawn-dusk meridian while for ions the region of highest flux is cusp shaped with a limited extent in both CGL and in MLT about noon. For both species this precipitation is associated with the precipitation of particles with approximately magnetosheath type energies associated with particles in the cusp and the various boundary regions in the magnetosphere.

Unlike for the case of the integral energy flux, there is no place in the oval where the precipitating ion number flux is a significant fraction of the electron flux. This is shown in Fig. 4 where the integral number flux is plotted as in Fig. 3. The figure illustrates that for all latitudes along this meridian the ion integral number flux is typically only approximately 1 percent or less of that for electrons. This shows that the precipitating ions do not play a role in general as the current carrier for the field aligned currents.

The global patterns of average energy for the electrons and ions show a similar

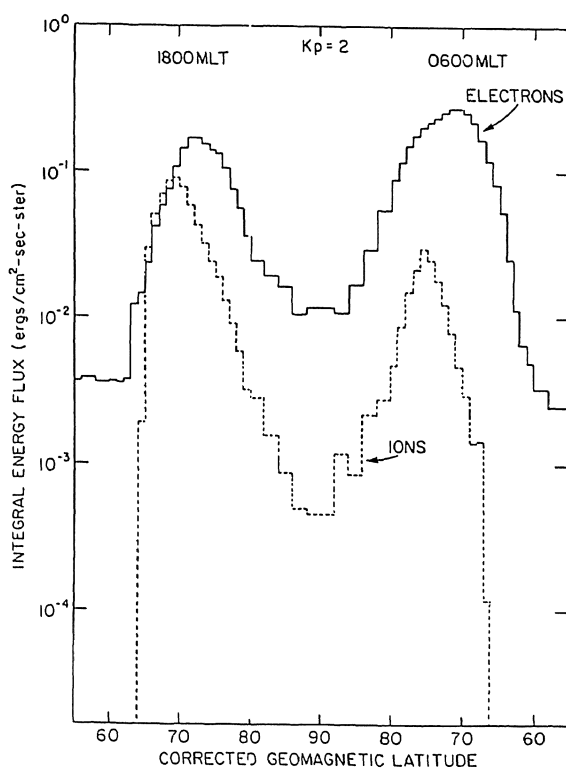


Fig. 3. The variation in the average value of the integral energy flux for precipitating electrons and ions along the dawn dusk meridian for $Kp = 2$.

symmetry to that for the integral energy flux for reflection about the dawn-dusk meridian. For the ions the highest average energies are seen on the dusk side of the oval with a clear minimum in the average energy in the oval in the pre-noon sector. The electrons by contrast show the highest average energies on the morningside of the oval with the oval minimum in the post-noon sector. This is seen more clearly in Fig. 5 where the average energy for electrons and ions is plotted along the dawn-dusk meridian as in Fig. 3. The highest average energies for electrons (ions) appear on the morning (evening) side of the oval equatorward of the peak in the ion (electron) average energy. Again this most likely reflects the paths followed by hot electrons and ions in the inner magnetosphere and is qualitatively in agreement with models for such transport (EJRI, 1978).

The two patterns are also similar in that they display a clear average energy minimum on the dayside along the noon meridian poleward of the region of maximum integral number flux. We interpret this as the signature of the particle energy dispersion produced by the motion of particles in the convection electric field after they have gained access to the cusp as has been reported for case studies by other researchers (REIFF *et al.*, 1977; SHELLY, 1979; BURCH *et al.*, 1980).

The principal difference in the two patterns is that the ions are much hotter than the electrons. The highest ion average energy is approximately 30 keV while that for electrons is

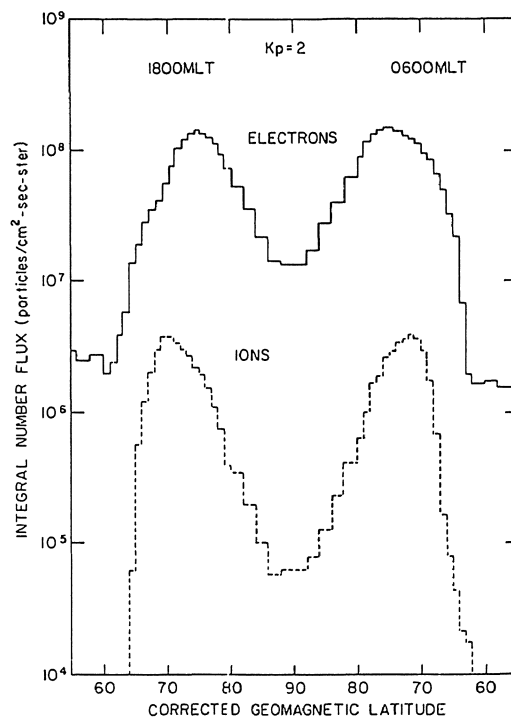


Fig. 4. The variation in the average value of the integral number flux for precipitating electrons and ions along the dawn dusk meridian for $Kp = 2$.

only 5 keV. The higher ion average energies explain why the ions can make a significant contribution to the integral energy flux even though their integral number flux is so low.

The patterns of height integrated Hall and Pedersen conductivities from precipitating electrons are similar to that for the electron integral energy flux. For both conductivities the highest values are observed on the evening side of the oval in a crescent shaped region symmetric about a meridian running post-midnight and post-noon. Conductivities in the cusp and polar cap are uniformly low on the order of .5 mhos or less. The general pattern of the Hall and Pedersen conductivities are the same except that the Hall conductivity exceeds the Pedersen by approximately a factor of 2. The patterns of integral number flux and integral energy flux for both electrons and ions and the pattern of the Hall and Pedersen conductivities for electrons all have been fit to simple functional forms to facilitate their use in global models (HARDY *et al.*, 1987; HARDY *et al.*, 1990).

The relative importance of the integral energy flux and number flux for electrons and ions is further illustrated in Table 1 where total input of these quantities spatially integrated over the hemisphere are listed. The hemispheric electron integral flux varies from 4.7 to 78 gigawatts while the ion's the variation is from 1.5 to 10 gigawatts. The ratio of the two is approximately constant at 15%. For the hemispheric integral number flux the electrons vary from 1.5×10^{25} to 10^{26} particles/second ster while the ions vary from 3.62×10^{23} to 1.5×10^{24} particles/second ster. The number flux ratio for the two species is less than 1.5%.

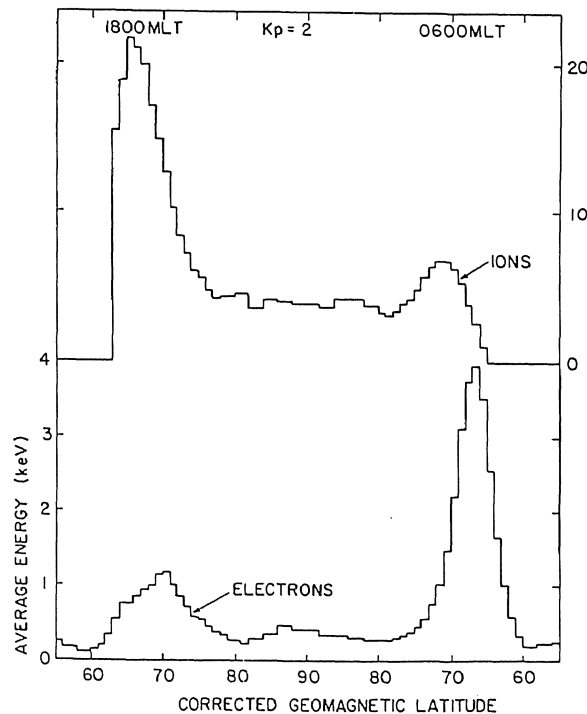


Fig. 5. The variation in the average values of the average energy for the precipitating electrons and ions along the dawn dusk meridian.

Table 1.

Kp	Electron energy flux (GW)	Electron number flux $\times 10^{25} \text{ (s-sr)}^{-1}$	Ion energy flux (GW)	Ion number flux $\times 10^{23} \text{ (s-sr)}^{-1}$
0	4.68	1.53	0.73	3.62
1	8.14	1.85	1.38	3.80
2	14.90	2.35	2.03	4.54
3	25.90	3.08	2.92	5.53
4	33.60	4.05	4.10	7.10
5	43.20	5.83	5.47	8.74
>6-	77.60	10.20	10.00	15.00

3.2 Auroral electron and ion precipitation relationship to interplanetary medium conditions

The spatial patterns in the integral energy flux and integral number flux maps generated in this study were similar to those for the Kp separation; only the quantitative details differed. For this reason in this study, we concentrated on the total hemispheric energy and number flux for the precipitating ions and electrons and their dependence on the solar wind speed and the IMF B_z component. We take the integrated hemispheric energy and number flux inputs (HEFLUX and HNFLUX) to be a measure of the character and strength of the solar wind-magnetosphere coupling. The integral number and energy flux as determined by integrating the extrapolated average spectra over the spherical area of the auroral region, are listed in Table 2 and plotted in Figs. 6 and 7. The fluxes determined from the unextrapolated spectra are also included in parentheses in Table 1. The integral values are plotted against B_z , for fixed V_{sw} , in the figures. Values from four of the initial thirty maps were omitted from the analyses due to their poor statistics.

The electron and ion HNFLUX and ion HEFLUX vary similarly with B_z , exhibiting a minimum value for weak to moderate B_z positive with symmetric increases for B_z greater (as far as we can tell) and less than the minimum. The electron HEFLUX shows no such increase for strong B_z positive. For increasing B_z negative the electron and ion HEFLUX's show the largest overall increase, by a factor of 5.8 and 3.7 respectively, from the minimum to the maximum negative B_z bin. This occurs for the lowest V_{sw} bin, with the factors decreasing for higher V_{sw} . The electron and ion HEFLUX's show smaller rises, both increasing by a factor of approximately 2.5, and with the factor remaining near this value for all V_{sw} . For B_z positive of the minimum all but the electron HEFLUX increase. The increase is largest for the ions with an average rise of a factor of 1.41 and 1.42 for the ion HEFLUX and HNFLUX respectively. The electron HNFLUX increases on average by a factor of 1.19. For fixed B_z values, the four hemispherically-integrated fluxes exhibit a near linear variation with V_{sw} . Again the electron and ion HEFLUX's show the largest variation increasing by an average factor of 4.08 and 2.11, respectively, over the full velocity range. The electron and ion HNFLUX's increase by a factor of 1.82 and 1.16, respectively.

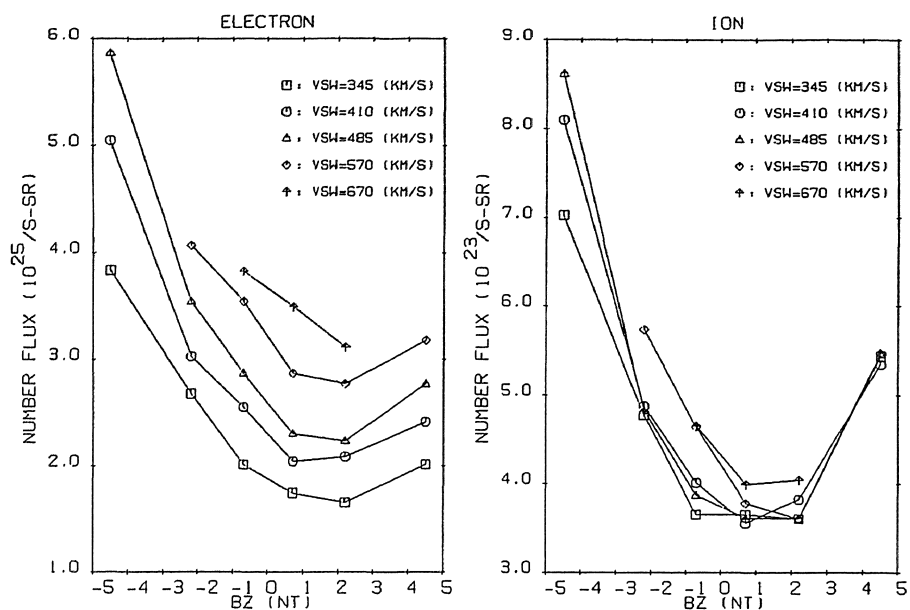


Fig. 6. The variation in the hemispheric integral number flux for electrons and ions plotted as a function of the magnitude of the IMF B_z component for five fixed ranges of the solar wind velocity.

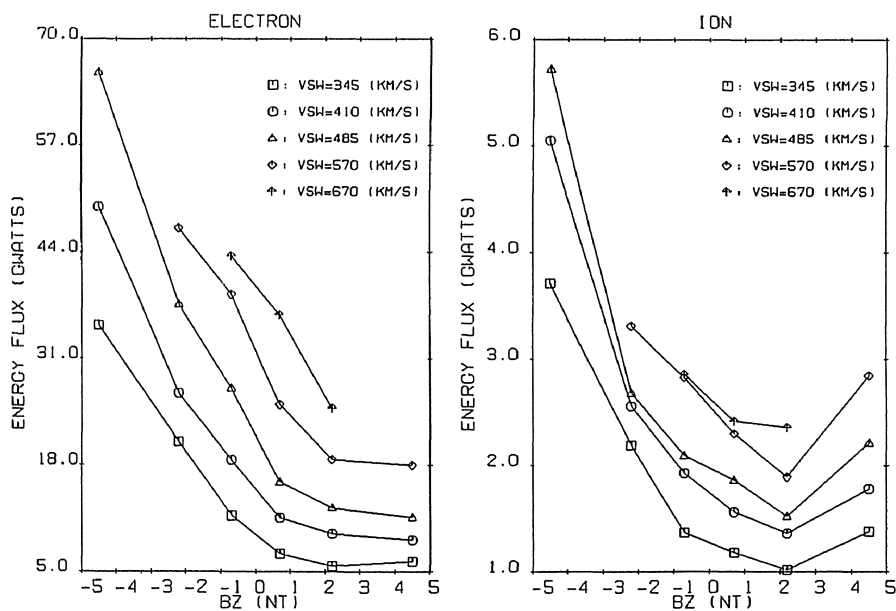


Fig. 7. The variation in the hemispheric integral energy flux for electrons and ions plotted as a function of the magnitude of the IMF B_z component for five fixed ranges of the solar wind velocity.

Table 2. Hemispheric fluxes (electron and ion).

B_z , nT	V_{sw} , km/s	Electron	Electron	Ion	Ion
		$H_{E_{FLUX}}$, GW	$H_{N_{FLUX}}$, $\times 10^{25} \text{ (s-sr)}^{-1}$	$H_{E_{FLUX}}$, GW	$H_{N_{FLUX}}$, $\times 10^{23} \text{ (s-sr)}^{-1}$
-4.5	346.	35.00 (34.50)	3.83 (3.83)	3.71 (2.63)	7.03 (6.61)
-2.2	346.	20.80 (20.50)	2.68 (2.68)	2.19 (1.77)	4.78 (4.61)
-0.7	346.	11.80 (11.70)	2.01 (2.01)	1.37 (1.20)	3.65 (3.58)
0.7	346.	7.18 (7.17)	1.74 (1.74)	1.18 (1.05)	3.65 (3.60)
2.2	346.	5.71 (5.70)	1.65 (1.65)	1.02 (0.96)	3.60 (3.58)
4.5	346.	6.23 (6.17)	2.01 (2.01)	1.38 (1.28)	5.44 (5.39)
-4.5	408.	49.50 (48.50)	5.05 (5.04)	5.05 (3.16)	8.10 (7.39)
-2.2	408.	26.80 (26.40)	3.03 (3.03)	2.56 (1.83)	4.88 (4.60)
-0.7	408.	18.60 (18.40)	2.55 (2.55)	1.93 (1.49)	4.01 (3.84)
0.7	408.	11.50 (11.40)	2.04 (2.04)	1.56 (1.23)	3.55 (3.42)
2.2	408.	9.58 (9.54)	2.08 (2.08)	1.36 (1.19)	3.82 (3.75)
4.5	408.	8.82 (8.79)	2.41 (2.41)	1.78 (1.55)	5.35 (5.26)
-4.5	485.	65.90 (63.30)	5.87 (5.86)	5.72 (3.21)	8.62 (7.71)
-2.2	485.	37.60 (36.50)	3.54 (3.54)	2.68 (1.76)	4.83 (4.48)
-0.7	485.	27.40 (26.90)	2.87 (2.87)	2.09 (1.45)	3.87 (3.62)
0.7	485.	15.90 (15.80)	2.30 (2.30)	1.86 (1.30)	3.61 (3.40)
2.2	485.	12.80 (12.70)	2.23 (2.23)	1.52 (1.23)	3.60 (3.48)
4.5	485.	11.60 (11.50)	2.77 (2.77)	2.21 (1.79)	5.46 (5.30)
-4.5	572.	68.40 (64.60)	5.93 (5.92)	5.70 (3.05)	8.53 (7.58)
-2.2	572.	46.90 (44.70)	4.07 (4.06)	3.31 (1.98)	5.74 (5.25)
-0.7	572.	38.80 (37.40)	3.54 (3.54)	2.83 (1.65)	4.64 (4.21)
0.7	572.	25.40 (24.70)	2.87 (2.86)	2.30 (1.41)	3.78 (3.45)
2.2	572.	18.70 (18.30)	2.77 (2.77)	1.89 (1.32)	3.60 (3.38)
4.5	572.	18.00 (17.80)	3.18 (3.18)	2.85 (1.95)	5.48 (5.15)
-4.5	677.	65.50 (60.00)	5.42 (5.40)	4.61 (2.27)	6.64 (5.86)
-2.2	677.	60.20 (55.20)	4.66 (4.64)	3.28 (1.89)	5.64 (5.13)
-0.7	677.	43.50 (41.10)	3.82 (3.81)	2.86 (1.64)	4.66 (4.22)
0.7	677.	36.30 (34.70)	3.49 (3.48)	2.42 (1.42)	3.99 (3.63)
2.2	677.	25.00 (24.30)	3.11 (3.11)	2.36 (1.43)	4.04 (3.70)
4.5	677.	20.30 (19.80)	3.13 (3.13)	2.76 (1.77)	4.85 (4.49)

The similarity in the variation of the four quantities with B_z and V_{sw} suggests that the variation can be fit by a relatively simple functional form. Two different functional forms were fit to the data. The two fits were of the forms;

$$f(B_z, V_{sw}) = a(B_z - b)^2 + cV_{sw} + d \quad (1)$$

Table 3. Hemispheric flux fit coefficients.

Coefficient*	Electrons		Ions	
	Energy flux	Number flux	Energy flux	Number flux
Method 1				
$a, x/nT^2$	0.59	0.06	0.07	0.14
b, nT	4.45	2.33	2.00	0.96
$c, x/(km/s)$	0.092	0.005	0.004	0.002
d, x	-34.20	-0.31	-0.38	2.66
$-d/c, km/s$	372.	62.	95.	-1330.
Method 2				
$a, l/nT$	0.08	0.03	0.06	0.04
b, nT	3.31	2.00	1.88	0.95
$c, x/(km/s)$	0.041	0.004	0.004	0.001
d, x	-8.65	-0.21	-0.32	3.29
$-d/c, km/s$	211.	-53.	80.	-3290.

*Variable x is in gigawatts (energy flux) or $(s-sr)^{-1}$ (number flux).

and

$$f(B_z, V_{sw}) = \left[a(B_z - b)^2 + 1 \right] [cV_{sw} + d] \quad (2)$$

These relations show that the data can be simply described as a family of parabolas whose minima are located at $B_z = b$ and are offset by a linear function of V_{sw} . The values of the four fit parameters for both techniques are listed in Table 3. A principal point of note is that these functional forms differ significantly from those used previously for solar wind-magnetosphere interactions (BAKER *et al.*, 1986 and references therein; BERGATZE *et al.*, 1985)

Although the final functional forms for both methods, given as Eqs. (1) and (2), have four free parameters each, the functional forms differ. Eq. (1) is the simpler, containing only the terms B_z^2 , B_z , and V_{sw} . Eq. (2) contains the additional cross terms, $V_{sw}B_z$ and $V_{sw}B_z^2$. Because there are four free parameters in each method the quality of fit should be approximately the same. A quantitative comparison of all data points to the calculated values using Eqs. (1) and (2) and the constants in Table 3 show the following: Method 1 resulted in 71% (94%) of the hemispheric fluxes being fit to within 10% (20%) of the data. The average correlation coefficient of a linear regression run between the data and fitted points for each of the four types of fluxes was 0.95. Method 2 yielded slightly better fits, with 80% (99%) of fitted points within 10% (20%) of the data. The average correlation coefficient in this case was 0.98.

3.3 Current structures and the electron and ion precipitation patterns

The study of the delta B 's and their relationship to the precipitation patterns provide important insights into processes throughout the oval. In this section, however, we concentrate on the interrelationship of these quantities in the dayside oval with special emphasis on the behavior in the cusp/cleft region. This is shown in Fig. 8 where we have plotted for the dayside oval the delta B vectors overlaid on color spectrograms of the electron

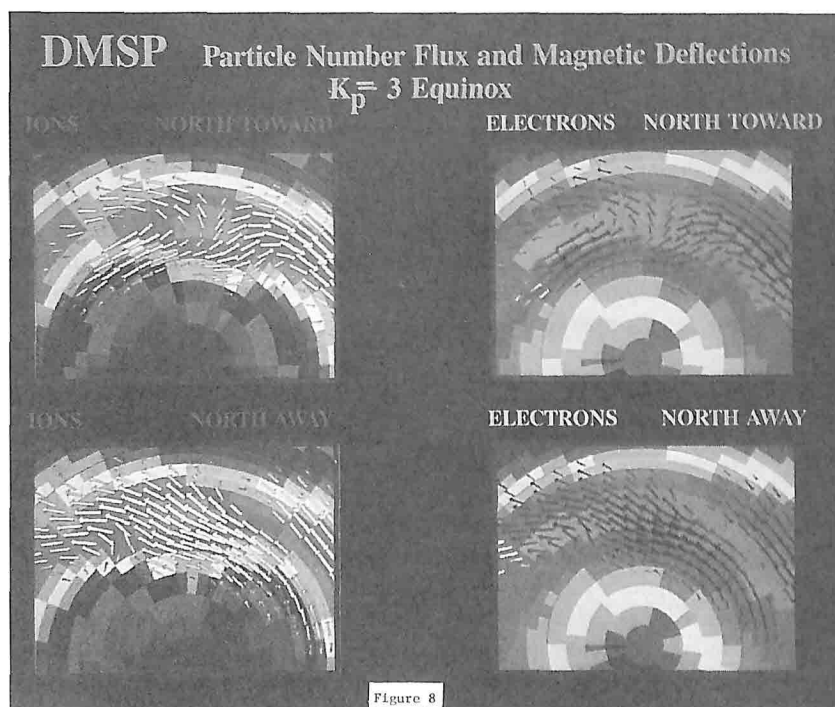


Fig. 8. Color spectrogram of the pattern of integral number flux for electrons and ions for $K_p = 3$ for IMF towards and away sector in the same format as Fig. 1. Superimposed on the figures as arrows are the vectors of the average δB 's for the same K_p and IMF conditions.

and ion integral number flux for the equinox season, for $K_p = 3$ and separately for the IMF towards and away sectors. The plots generally represent behavior seen at all other levels of K_p for the same season and sector separation. Significantly different behavior is seen for summer and winter solstice conditions but will not be presented here.

The principal observation here is the change in the precipitation pattern and more radically in the δB configuration with the change in sector structure. For the north pole toward sector case, the δB vectors observed in the pre-noon sector show there is a single pair of field aligned current sheets we identify with the dayside region-1/region-2 system (IJIIMA and POTEIRA, 1976). On the post-noon side the δB 's also indicate a region-1/region-2 system but in addition they indicate the presence of a second pair of field aligned currents associated exclusively with the cusp/cleft precipitation (REIFF and BURCH, 1985). For the north pole away sector case the δB pattern changes markedly with only the region-1/region-2 current post-noon and the region-1/region-2 currents and cusp currents seen pre-noon. The particle precipitation patterns by contrast do not show such an extreme change with sector with only a slight shift of the pattern pre-noon for the towards case and post-noon for the away case. The cusp current system for the two cases extends in MLT four to five hours from noon post-noon and pre-noon for the two cases, respectively. For both cases the δB vectors diverge through the region of maximum ion integral flux that we identify

with the cusp. This divergence is not associated with any marked feature in the electron precipitation pattern.

What is most significant is that the precipitation pattern varies so little with the sector structure. Thus for the north pole towards case in the pre-noon sector the region-1 currents near noon collocate with the region of relatively cold high intensity precipitation we take to originate in the boundary regions while in the post-noon sector the cusp currents are embedded in regions with the same characteristics for the electron and ion precipitation. Similarly for the north pole away sector case the pattern is reversed with the post-noon region-1 and the pre-noon cusp currents embedded in the same plasma regime. In addition, in the sector with both the region-1/region-2 currents and the cusp current the region-1 currents are seen to lie mostly in the region of cold cusp/cleft like precipitation.

4. Summary and Conclusions

We have reviewed here a series of studies conducted to determine the average pattern of auroral precipitation, electron induced Hall and Pedersen conductivities and delta B vectors as a function of the level of geomagnetic activity, conditions in the interplanetary medium and season. For all quantities studied and with all separations used the resulting patterns show smooth and regular variations with MLT, CGL and the sort parameters used. We contend that these results accurately depict the underlying pattern within the auroral zone of the quantities studied and their interrelationship. We maintain this despite the fact that there are processes such as substorms that occur on time scales short compared to those inherent in our global determinations and phenomena such as discrete auroral arc that occur on spatial scales short compared to the scale of our latitudinal divisions. These studies indicate that such processes are perturbations to the processes producing these underlying patterns. As such these studies provide important insights into the processes taking place in the auroral zone and the interconnection of the auroral zone to other regions in the magnetosphere and to the conditions in the interplanetary medium. In addition these results provide important boundary conditions to models that attempt to globally model the auroral zone and magnetosphere.

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